LAB TASK - 09

Write a program to implement Floyd Warshall Algorithm to find the shortest path between every pair of the vertices.

CODE:

```
#include <iostream>
using namespace std;

#define INF 99999

int main() {
   int n;
   cout << "Enter the number of vertices: ";
   cin >> n;

   int dist[100][100];

   cout << "Enter the adjacency matrix (use 99999 for INF):\n";
   for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++) {
         cin >> dist[i][j];
      }
   }

   for (int k = 0; k < n; k++) {
      for (int i = 0; i < n; i++) {
         for (int j = 0; j < n; j++) {
            if (dist[i][k] != INF && dist[k][j] != INF &&
               dist[i][k] + dist[k][j] < dist[i][j]) {
               dist[i][j] = dist[i][k] + dist[k][j];
            }
         }
      }
```

```
    }

    cout << "\nShortest distance matrix:\n";
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (dist[i][j] == INF)
                cout << "INF\t";
            else
                cout << dist[i][j] << "\t";
        }
        cout << endl;
    }

    return 0;
}
```

OUTPUT:

```
god@god-HP-Laptop-14s-er0xxx:~/daa/260CT$ ./p1
Enter the number of vertices: 4
Enter the adjacency matrix (use 99999 for INF):
0 5 99999 10
99999 0 3 99999
99999 99999 0 1
99999 99999 99999 0

Shortest distance matrix:
0       5       8       9
INF     0       3       4
INF     INF     0       1
INF     INF     INF     0
god@god-HP-Laptop-14s-er0xxx:~/daa/260CT$
```

Time complexity: $O(V^3)$

Space complexity: $O(V^2)$